

Exact values



1

a $\frac{\sqrt{3}}{2}$

b $\frac{1}{2}$

c $\frac{1}{\sqrt{3}}$

d $\frac{1}{\sqrt{2}}$

2

a $\theta = 30^\circ$

b $\cos \theta = \frac{\sqrt{3}}{2}$

3

a $x = \frac{\sqrt{30}}{2}$

b $h = 1$

c $x = 2$

d $P = 17$

e $w = 22$

f $x = 4\sqrt{6}, y = 4\sqrt{3}$

g $u = 8, v = 4\sqrt{3}$

h $a = \frac{9\sqrt{3}}{2}, b = \frac{9}{2}$

i $x = 7\sqrt{2}, y = 14\sqrt{2}$

j $a = 9\sqrt{2}, b = 6\sqrt{3}, c = 12\sqrt{3}$

4

a 45°

b 60°

5

a $\angle DCB, \angle BAC$

b $m = \frac{20\sqrt{21}}{3}, n = 5\sqrt{21}$

6

a i $p = \frac{5\sqrt{3}}{3}$

ii $q = \frac{10\sqrt{3}}{3}$

iii $r = \frac{5}{2}$

iv $s = \frac{5\sqrt{3}}{2}$

b $\frac{15\sqrt{3}}{2} + \frac{5}{2}$

7

a $x = 5$

b $y = 5\sqrt{3}$

c $5\sqrt{3} + 35$